CHAPTER 9

Plotting and Visualization

Making informative visualizations (sometimes called *plots*) is one of the most important tasks in data analysis. It may be a part of the exploratory process—for example, to help identify outliers or needed data transformations, or as a way of generating ideas for models. For others, building an interactive visualization for the web may be the end goal. Python has many add-on libraries for making static or dynamic visualizations, but I'll be mainly focused on matplotlib and libraries that build on top of it.

matplotlib is a desktop plotting package designed for creating (mostly two-dimensional) publication-quality plots. The project was started by John Hunter in 2002 to enable a MATLAB-like plotting interface in Python. The matplotlib and IPython communities have collaborated to simplify interactive plotting from the IPython shell (and now, Jupyter notebook). matplotlib supports various GUI backends on all operating systems and additionally can export visualizations to all of the common vector and raster graphics formats (PDF, SVG, JPG, PNG, BMP, GIF, etc.). With the exception of a few diagrams, nearly all of the graphics in this book were produced using matplotlib.

Over time, matplotlib has spawned a number of add-on toolkits for data visualization that use matplotlib for their underlying plotting. One of these is seaborn, which we explore later in this chapter.

The simplest way to follow the code examples in the chapter is to use interactive plotting in the Jupyter notebook. To set this up, execute the following statement in a Jupyter notebook:

```
%matplotlib notebook
```

9.1 A Brief matplotlib API Primer

With matplotlib, we use the following import convention:

```
In [11]: import matplotlib.pyplot as plt
```

After running `%matplotlib notebook` in Jupyter (or simply `%matplotlib` in IPython), we can try creating a simple plot. If everything is set up right, a line plot like Figure 9-1 should appear:

```
In [12]: import numpy as np

In [13]: data = np.arange(10)

In [14]: data
Out[14]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])

In [15]: plt.plot(data)
```

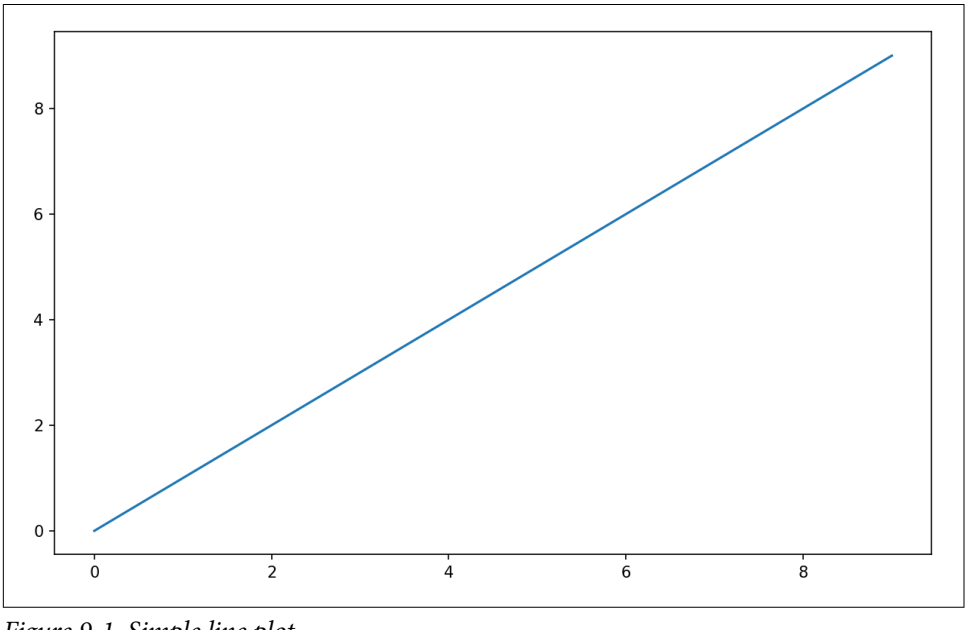


Figure 9-1. Simple line plot

While libraries like seaborn and pandas's built-in plotting functions will deal with many of the mundane details of making plots, should you wish to customize them beyond the function options provided, you will need to learn a bit about the matplotlib API.



There is not enough room in the book to give a comprehensive treatment to the breadth and depth of functionality in matplotlib. It should be enough to teach you the ropes to get up and running. The matplotlib gallery and documentation are the best resource for learning advanced features.

Figures and Subplots

Plots in matplotlib reside within a `Figure` object. You can create a new figure with `plt.figure`:

```
In [16]: fig = plt.figure()
```

In IPython, an empty plot window will appear, but in Jupyter nothing will be shown until we use a few more commands. `plt.figure` has a number of options; notably, `figsize` will guarantee the figure has a certain size and aspect ratio if saved to disk.

You can't make a plot with a blank figure. You have to create one or more `subplots` using `add_subplot`:

```
In [17]: ax1 = fig.add_subplot(2, 2, 1)
```

This means that the figure should be 2 × 2 (so up to four plots in total), and we're selecting the first of four subplots (numbered from 1). If you create the next two subplots, you'll end up with a visualization that looks like Figure 9-2:

```
In [18]: ax2 = fig.add_subplot(2, 2, 2)

In [19]: ax3 = fig.add_subplot(2, 2, 3)
```

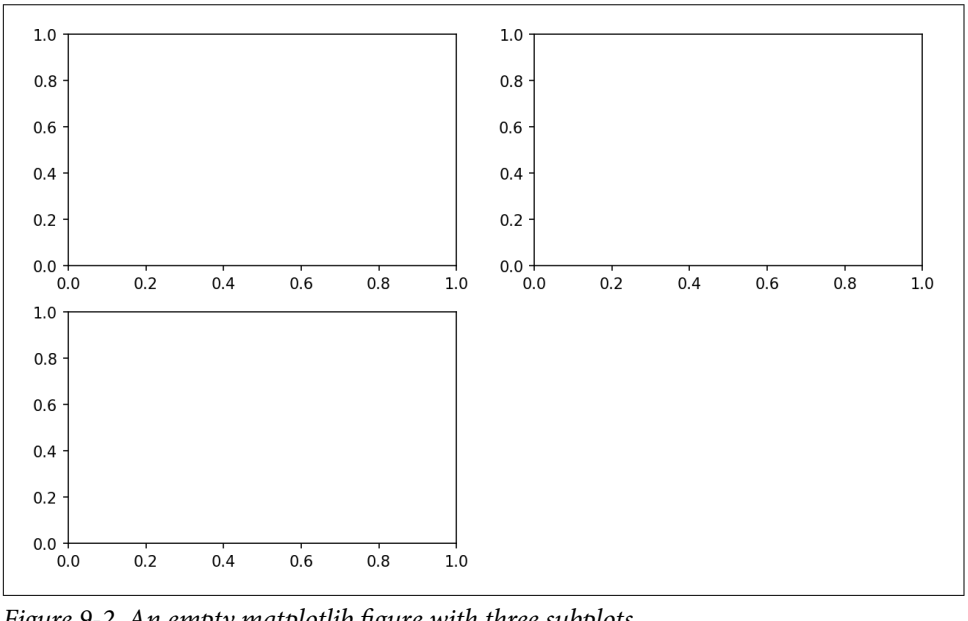


Figure 9-2. An empty matplotlib figure with three subplots



One nuance of using Jupyter notebooks is that plots are reset after each cell is evaluated, so for more complex plots you must put all of the plotting commands in a single notebook cell.

Here we run all of these commands in the same cell:

```
fig = plt.figure()
ax1 = fig.add_subplot(2, 2, 1)
ax2 = fig.add_subplot(2, 2, 2)
ax3 = fig.add_subplot(2, 2, 3)
```

When you issue a plotting command like `plt.plot([1.5, 3.5, -2, 1.6])`, matplotlib draws on the last figure and subplot used (creating one if necessary), thus hiding the figure and subplot creation. So if we add the following command, you'll get something like Figure 9-3:

```
In [20]: plt.plot(np.random.randn(50).cumsum(), 'k--')
```

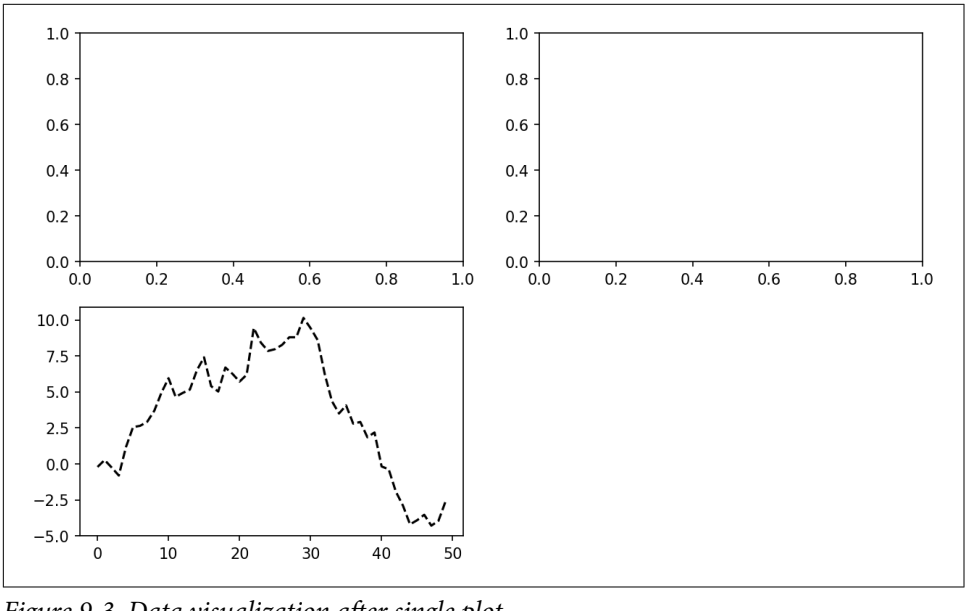


Figure 9-3. Data visualization after single plot

The `'k--'` is a *style* option instructing matplotlib to plot a black dashed line. The objects returned by `fig.add_subplot` here are `AxesSubplot` objects, on which you can directly plot on the other empty subplots by calling each one's instance method (see Figure 9-4):

```
In [21]: _ = ax1.hist(np.random.randn(100), bins=20, color='k', alpha=0.3)

In [22]: ax2.scatter(np.arange(30), np.arange(30) + 3 * np.random.randn(30))
```

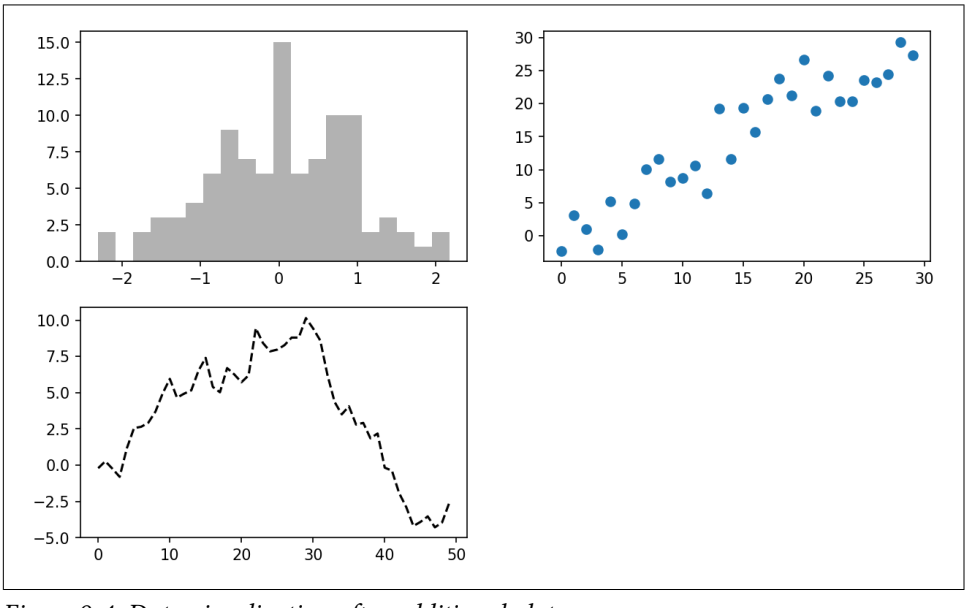


Figure 9-4. Data visualization after additional plots

You can find a comprehensive catalog of plot types in the matplotlib documentation.

Creating a figure with a grid of subplots is a very common task, so matplotlib includes a convenience method, `plt.subplots`, that creates a new figure and returns a NumPy array containing the created subplot objects:

```
In [24]: fig, axes = plt.subplots(2, 3)

In [25]: axes
Out[25]:
array([[<matplotlib.axes._subplots.AxesSubplot object at 0x7fb626374048>,
        <matplotlib.axes._subplots.AxesSubplot object at 0x7fb62625db00>,
        <matplotlib.axes._subplots.AxesSubplot object at 0x7fb6262f6c88>],
       [<matplotlib.axes._subplots.AxesSubplot object at 0x7fb6261a36a0>,
        <matplotlib.axes._subplots.AxesSubplot object at 0x7fb626181860>,
        <matplotlib.axes._subplots.AxesSubplot object at 0x7fb6260fd4e0>]], dtype
=object)
```

This is very useful, as the `axes` array can be easily indexed like a two-dimensional array; for example, `axes[0, 1]`. You can also indicate that subplots should have the same x- or y-axis using `sharex` and `sharey`, respectively. This is especially useful when you're comparing data on the same scale; otherwise, matplotlib autoscales plot limits independently. See Table 9-1 for more on this method.

Table 9-1. pyplot.subplots options

Argument	Description
`nrows`	Number of rows of subplots
`ncols`	Number of columns of subplots
`sharex`	All subplots should use the same x-axis ticks (adjusting the `xlim` will affect all subplots)
`sharey`	All subplots should use the same y-axis ticks (adjusting the `ylim` will affect all subplots)
`subplot_kw`	Dict of keywords passed to `add_subplot` call used to create each subplot
`**fig_kw`	Additional keywords to `subplots` are used when creating the figure, such as `plt.subplots(2, 2, figsize=(8, 6))`

Adjusting the spacing around subplots

By default matplotlib leaves a certain amount of padding around the outside of the subplots and spacing between subplots. This spacing is all specified relative to the height and width of the plot, so that if you resize the plot either programmatically or manually using the GUI window, the plot will dynamically adjust itself. You can change the spacing using the `subplots_adjust` method on `Figure` objects, also available as a top-level function:

```
subplots_adjust(left=None, bottom=None, right=None, top=None,
                wspace=None, hspace=None)
```

`wspace` and `hspace` controls the percent of the figure width and figure height, respectively, to use as spacing between subplots. Here is a small example where I shrink the spacing all the way to zero (see Figure 9-5):

```
fig, axes = plt.subplots(2, 2, sharex=True, sharey=True)
for i in range(2):
    for j in range(2):
        axes[i, j].hist(np.random.randn(500), bins=50, color='k', alpha=0.5)
plt.subplots_adjust(wspace=0, hspace=0)
```

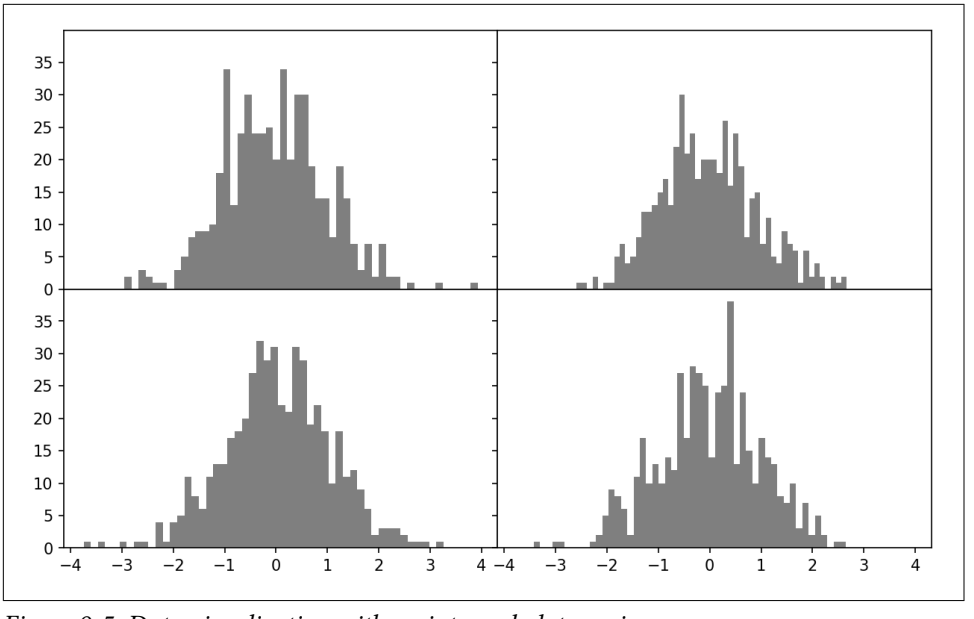


Figure 9-5. Data visualization with no inter-subplot spacing

You may notice that the axis labels overlap. matplotlib doesn't check whether the labels overlap, so in a case like this you would need to fix the labels yourself by specifying explicit tick locations and tick labels (we'll look at how to do this in the following sections).

Colors, Markers, and Line Styles

Matplotlib's main `plot` function accepts arrays of x and y coordinates and optionally a string abbreviation indicating color and line style. For example, to plot `x` versus `y` with green dashes, you would execute:

```
ax.plot(x, y, 'g--')
```

This way of specifying both color and line style in a string is provided as a convenience; in practice if you were creating plots programmatically you might prefer not to have to munge strings together to create plots with the desired style. The same plot could also have been expressed more explicitly as:

```
ax.plot(x, y, linestyle='--', color='g')
```

There are a number of color abbreviations provided for commonly used colors, but you can use any color on the spectrum by specifying its hex code (e.g., `'#CECECE'`). You can see the full set of line styles by looking at the docstring for `plot` (use `plot?` in IPython or Jupyter).

Line plots can additionally have *markers* to highlight the actual data points. Since matplotlib creates a continuous line plot, interpolating between points, it can occasionally be unclear where the points lie. The marker can be part of the style string, which must have color followed by marker type and line style (see Figure 9-6):

```
In [30]: from numpy.random import randn

In [31]: plt.plot(randn(30).cumsum(), 'ko--')
```

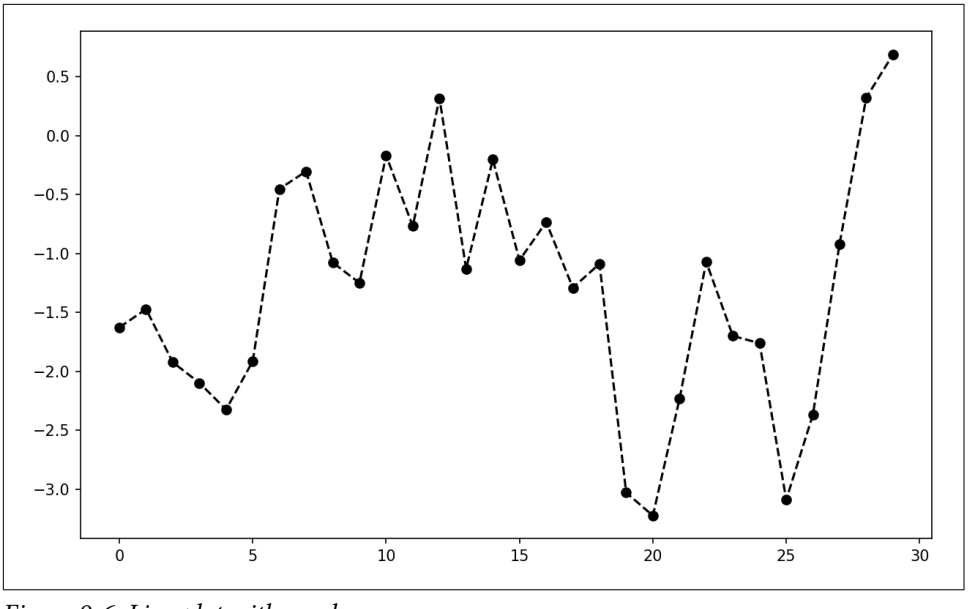


Figure 9-6. Line plot with markers

This could also have been written more explicitly as:

```
plot(randn(30).cumsum(), color='k', linestyle='dashed', marker='o')
```

For line plots, you will notice that subsequent points are linearly interpolated by default. This can be altered with the `drawstyle` option (Figure 9-7):

```
In [33]: data = np.random.randn(30).cumsum()

In [34]: plt.plot(data, 'k--', label='Default')
Out[34]: [<matplotlib.lines.Line2D at 0x7fb624d86160>]

In [35]: plt.plot(data, 'k-', drawstyle='steps-post', label='steps-post')
Out[35]: [<matplotlib.lines.Line2D at 0x7fb624d869e8>]

In [36]: plt.legend(loc='best')
```

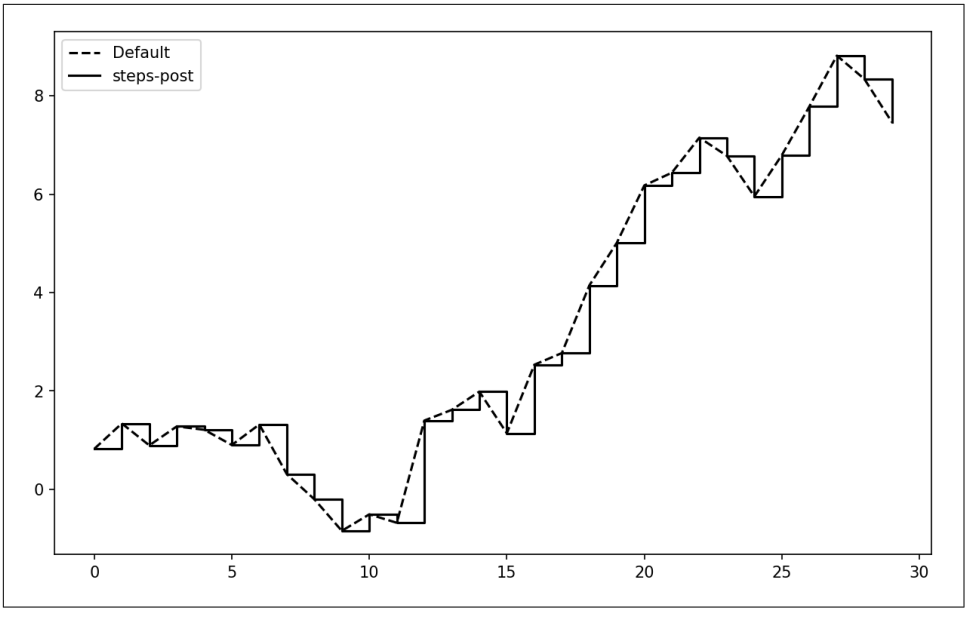


Figure 9-7. Line plot with different drawstyle options

You may notice output like `<matplotlib.lines.Line2D at ...>` when you run this. matplotlib returns objects that reference the plot subcomponent that was just added. A lot of the time you can safely ignore this output. Here, since we passed the `label` arguments to `plot`, we are able to create a plot legend to identify each line using `plt.legend`.



You must call `plt.legend` (or `ax.legend`, if you have a reference to the axes) to create the legend, whether or not you passed the `label` options when plotting the data.

Ticks, Labels, and Legends

For most kinds of plot decorations, there are two main ways to do things: using the procedural `pyplot` interface (i.e., `matplotlib.pyplot`) and the more object-oriented native matplotlib API.

The `pyplot` interface, designed for interactive use, consists of methods like `xlim`, `xticks`, and `xticklabels`. These control the plot range, tick locations, and tick labels, respectively. They can be used in two ways:

- Called with no arguments returns the current parameter value (e.g., `plt.xlim()` returns the current x-axis plotting range)
- Called with parameters sets the parameter value (e.g., `plt.xlim([0, 10])`, sets the x-axis range to 0 to 10)

All such methods act on the active or most recently created `AxesSubplot`. Each of them corresponds to two methods on the subplot object itself; in the case of `xlim` these are `ax.get_xlim` and `ax.set_xlim`. I prefer to use the subplot instance methods myself in the interest of being explicit (and especially when working with multiple subplots), but you can certainly use whichever you find more convenient.

Setting the title, axis labels, ticks, and ticklabels

To illustrate customizing the axes, I'll create a simple figure and plot of a random walk (see Figure 9-8):

```
In [37]: fig = plt.figure()

In [38]: ax = fig.add_subplot(1, 1, 1)

In [39]: ax.plot(np.random.randn(1000).cumsum())
```

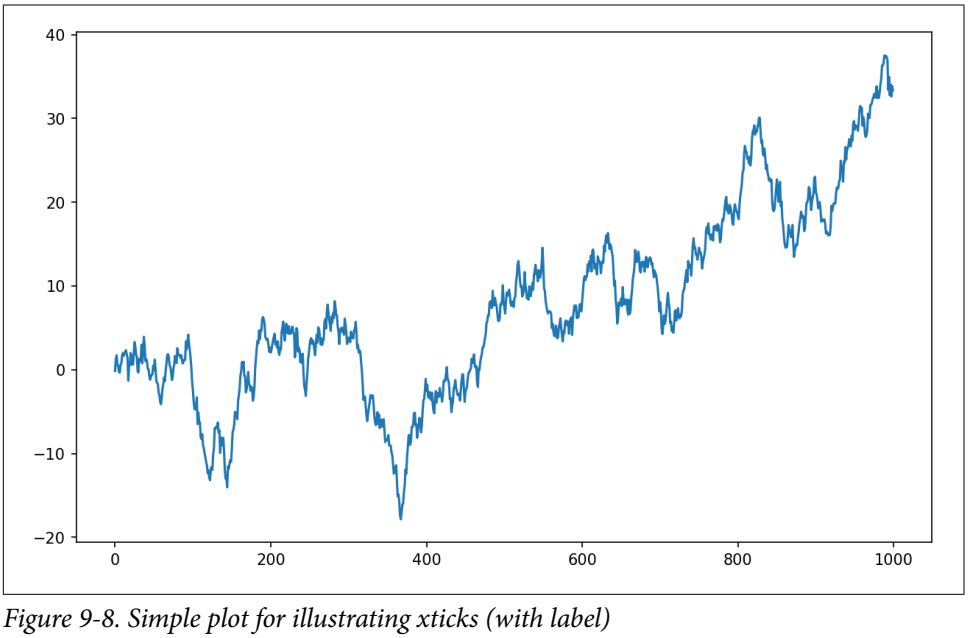


Figure 9-8. Simple plot for illustrating xticks (with label)

To change the x-axis ticks, it's easiest to use `set_xticks` and `set_xticklabels`. The former instructs matplotlib where to place the ticks along the data range; by default

these locations will also be the labels. But we can set any other values as the labels using `set_xticklabels`:

```
In [40]: ticks = ax.set_xticks([0, 250, 500, 750, 1000])

In [41]: labels = ax.set_xticklabels(['one', 'two', 'three', 'four', 'five'],
    ....:                            rotation=30, fontsize='small')
```

The `rotation` option sets the x tick labels at a 30-degree rotation. Lastly, `set_xlabel` gives a name to the x-axis and `set_title` the subplot title (see Figure 9-9 for the resulting figure):

```
In [42]: ax.set_title('My first matplotlib plot')
Out[42]: <matplotlib.text.Text at 0x7fb624d055f8>

In [43]: ax.set_xlabel('Stages')
```

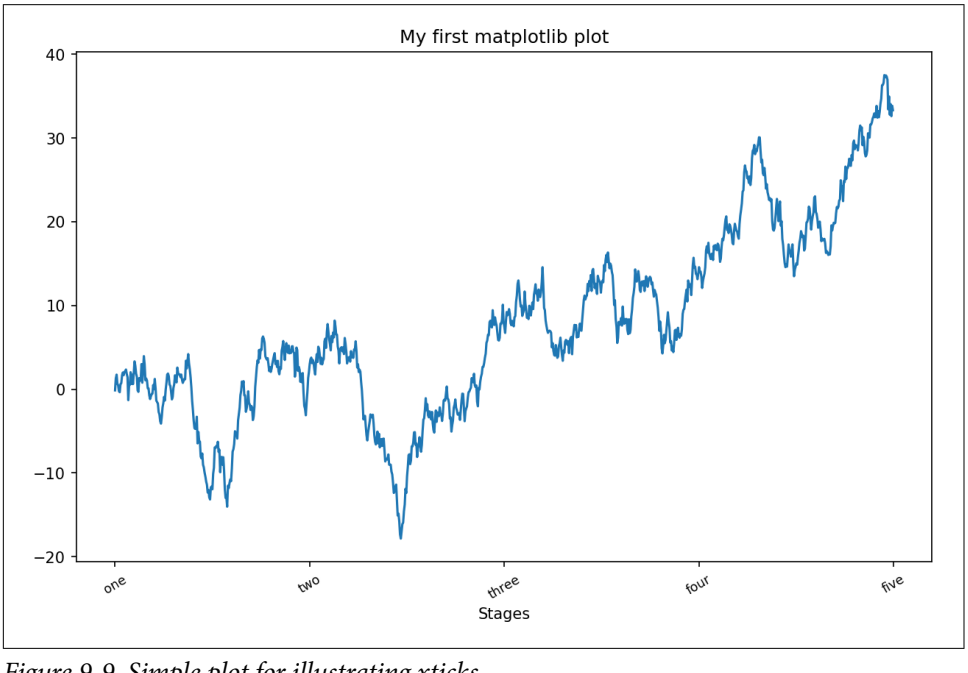


Figure 9-9. Simple plot for illustrating xticks

Modifying the y-axis consists of the same process, substituting `y` for `x` in the above. The axes class has a `set` method that allows batch setting of plot properties. From the prior example, we could also have written:

```
props = {
    'title': 'My first matplotlib plot',
    'xlabel': 'Stages'
}
ax.set(**props)
```

Adding legends

Legends are another critical element for identifying plot elements. There are a couple of ways to add one. The easiest is to pass the `label` argument when adding each piece of the plot:

```
In [44]: from numpy.random import randn

In [45]: fig = plt.figure(); ax = fig.add_subplot(1, 1, 1)

In [46]: ax.plot(randn(1000).cumsum(), 'k', label='one')
Out[46]: [<matplotlib.lines.Line2D at 0x7fb624bdf860>]

In [47]: ax.plot(randn(1000).cumsum(), 'k--', label='two')
Out[47]: [<matplotlib.lines.Line2D at 0x7fb624be90f0>]

In [48]: ax.plot(randn(1000).cumsum(), 'k.', label='three')
Out[48]: [<matplotlib.lines.Line2D at 0x7fb624be9160>]
```

Once you've done this, you can either call `ax.legend()` or `plt.legend()` to automatically create a legend. The resulting plot is in Figure 9-10:

```
In [49]: ax.legend(loc='best')
```

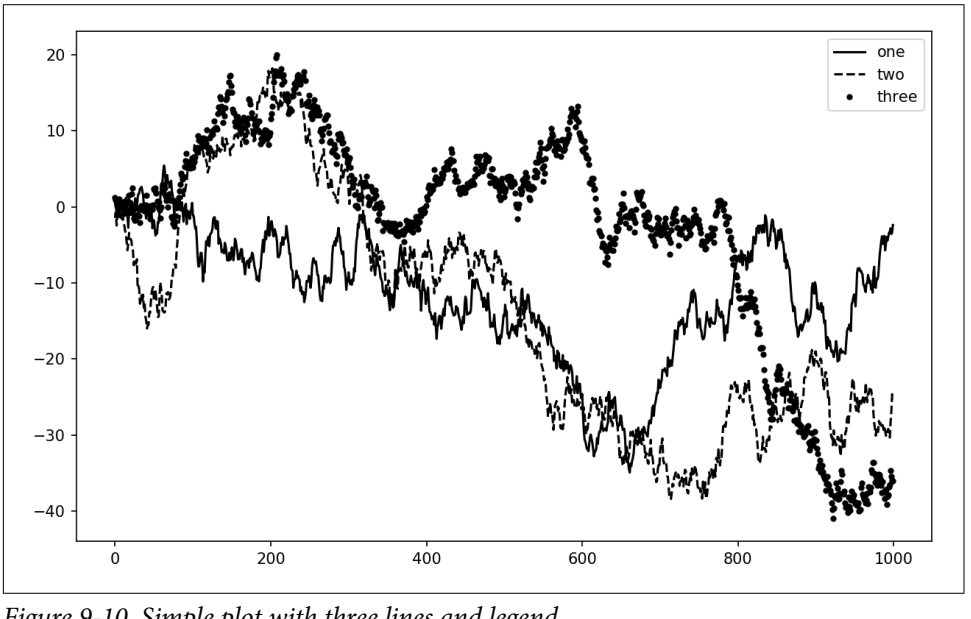


Figure 9-10. Simple plot with three lines and legend

The `legend` method has several other choices for the location `loc` argument. See the docstring (with `ax.legend?`) for more information.

The `loc` tells matplotlib where to place the plot. If you aren't picky, `'best'` is a good option, as it will choose a location that is most out of the way. To exclude one or more elements from the legend, pass no label or `label='_nolegend_'`.

Annotations and Drawing on a Subplot

In addition to the standard plot types, you may wish to draw your own plot annotations, which could consist of text, arrows, or other shapes. You can add annotations and text using the `text`, `arrow`, and `annotate` functions. `text` draws text at given coordinates `(x, y)` on the plot with optional custom styling:

```
ax.text(x, y, 'Hello world!',
        family='monospace', fontsize=10)
```

Annotations can draw both text and arrows arranged appropriately. As an example, let's plot the closing S&P 500 index price since 2007 (obtained from Yahoo! Finance) and annotate it with some of the important dates from the 2008–2009 financial crisis. You can most easily reproduce this code example in a single cell in a Jupyter notebook. See Figure 9-11 for the result:

```
from datetime import datetime

fig = plt.figure()
ax = fig.add_subplot(1, 1, 1)

data = pd.read_csv('examples/spx.csv', index_col=0, parse_dates=True)
spx = data['SPX']

spx.plot(ax=ax, style='k-')

crisis_data = [
    (datetime(2007, 10, 11), 'Peak of bull market'),
    (datetime(2008, 3, 12), 'Bear Stearns Fails'),
    (datetime(2008, 9, 15), 'Lehman Bankruptcy')
]

for date, label in crisis_data:
    ax.annotate(label, xy=(date, spx.asof(date) + 75),
                xytext=(date, spx.asof(date) + 225),
                arrowprops=dict(facecolor='black', headwidth=4, width=2,
                                headlength=4),
                horizontalalignment='left', verticalalignment='top')

# Zoom in on 2007-2010
ax.set_xlim(['1/1/2007', '1/1/2011'])
ax.set_ylim([600, 1800])

ax.set_title('Important dates in the 2008-2009 financial crisis')
```

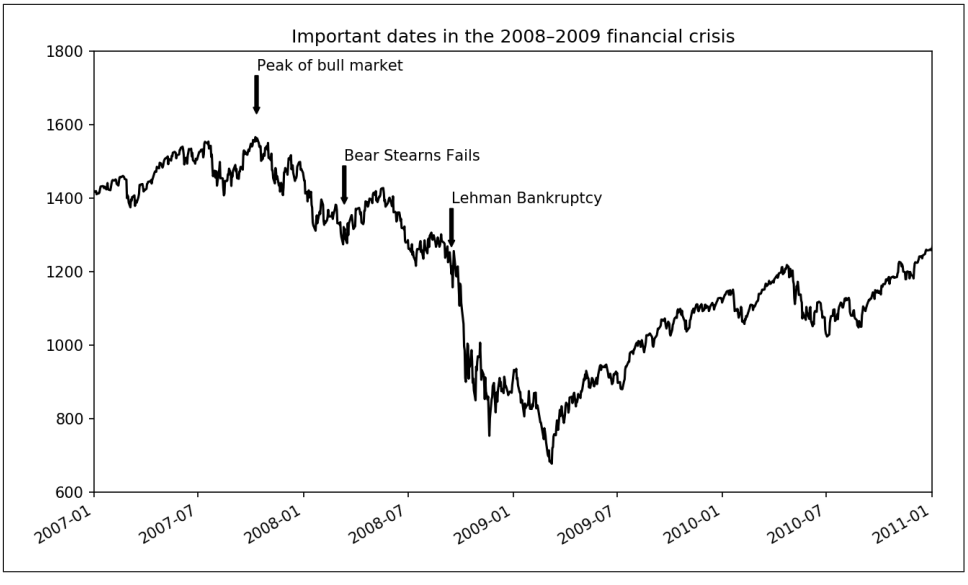


Figure 9-11. Important dates in the 2008–2009 financial crisis

There are a couple of important points to highlight in this plot: the `ax.annotate` method can draw labels at the indicated x and y coordinates. We use the `set_xlim` and `set_ylim` methods to manually set the start and end boundaries for the plot rather than using matplotlib's default. Lastly, `ax.set_title` adds a main title to the plot.

See the online matplotlib gallery for many more annotation examples to learn from.

Drawing shapes requires some more care. matplotlib has objects that represent many common shapes, referred to as *patches*. Some of these, like `Rectangle` and `Circle`, are found in `matplotlib.pyplot`, but the full set is located in `matplotlib.patches`.

To add a shape to a plot, you create the patch object `shp` and add it to a subplot by calling `ax.add_patch(shp)` (see Figure 9-12):

```
fig = plt.figure()
ax = fig.add_subplot(1, 1, 1)

rect = plt.Rectangle((0.2, 0.75), 0.4, 0.15, color='k', alpha=0.3)
circ = plt.Circle((0.7, 0.2), 0.15, color='b', alpha=0.3)
pgon = plt.Polygon([[0.15, 0.15], [0.35, 0.4], [0.2, 0.6]],
                   color='g', alpha=0.5)

ax.add_patch(rect)
ax.add_patch(circ)
ax.add_patch(pgon)
```

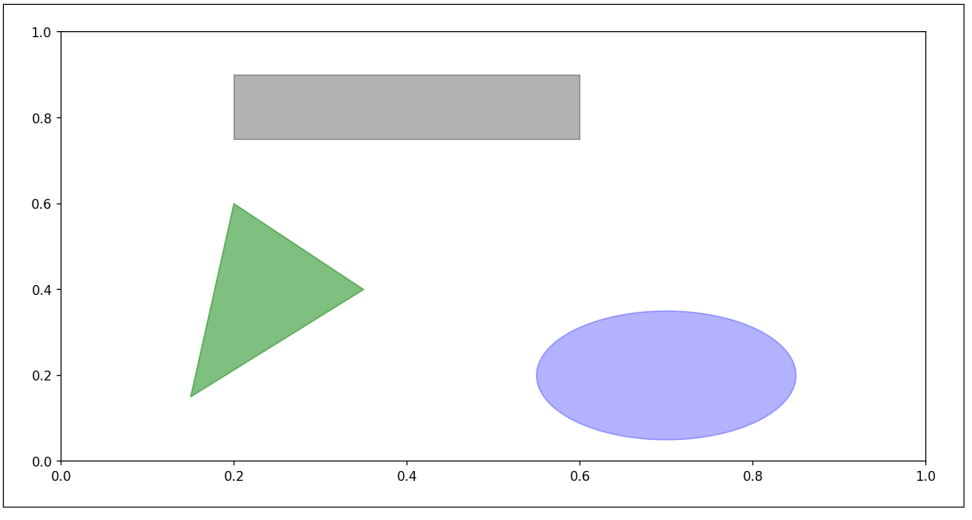


Figure 9-12. Data visualization composed from three different patches

If you look at the implementation of many familiar plot types, you will see that they are assembled from patches.

Saving Plots to File

You can save the active figure to file using `plt.savefig`. This method is equivalent to the figure object's `savefig` instance method. For example, to save an SVG version of a figure, you need only type:

```
plt.savefig('figpath.svg')
```

The file type is inferred from the file extension. So if you used `.pdf` instead, you would get a PDF. There are a couple of important options that I use frequently for publishing graphics: `dpi`, which controls the dots-per-inch resolution, and `bbox_inches`, which can trim the whitespace around the actual figure. To get the same plot as a PNG with minimal whitespace around the plot and at 400 DPI, you would do:

```
plt.savefig('figpath.png', dpi=400, bbox_inches='tight')
```

`savefig` doesn't have to write to disk; it can also write to any file-like object, such as a `BytesIO`:

```
from io import BytesIO
buffer = BytesIO()
plt.savefig(buffer)
plot_data = buffer.getvalue()
```

See Table 9-2 for a list of some other options for `savefig`.

Table 9-2. Figure.savefig options

Argument	Description
`fname`	String containing a filepath or a Python file-like object. The figure format is inferred from the file extension (e.g., `.pdf` for PDF or `.png` for PNG)
`dpi`	The figure resolution in dots per inch; defaults to 100 out of the box but can be configured
`facecolor`, `edgecolor`	The color of the figure background outside of the subplots; `'w'` (white), by default
`format`	The explicit file format to use (`'png'`, `'pdf'`, `'svg'`, `'ps'`, `'eps'`, ...)
`bbox_inches`	The portion of the figure to save; if `'tight'` is passed, will attempt to trim the empty space around the figure

matplotlib Configuration

matplotlib comes configured with color schemes and defaults that are geared primarily toward preparing figures for publication. Fortunately, nearly all of the default behavior can be customized via an extensive set of global parameters governing figure size, subplot spacing, colors, font sizes, grid styles, and so on. One way to modify the configuration programmatically from Python is to use the `rc` method; for example, to set the global default figure size to be 10×10, you could enter:

```
plt.rc('figure', figsize=(10, 10))
```

The first argument to `rc` is the component you wish to customize, such as `'figure'`, `'axes'`, `'xtick'`, `'ytick'`, `'grid'`, `'legend'`, or many others. After that can follow a sequence of keyword arguments indicating the new parameters. An easy way to write down the options in your program is as a dict:

```
font_options = {'family' : 'monospace',
                'weight' : 'bold',
                'size'   : 'small'}
plt.rc('font', **font_options)
```

For more extensive customization and to see a list of all the options, matplotlib comes with a configuration file *matplotlibrc* in the *matplotlib/mpl-data* directory. If you customize this file and place it in your home directory titled *.matplotlibrc*, it will be loaded each time you use matplotlib.

As we'll see in the next section, the seaborn package has several built-in plot themes or *styles* that use matplotlib's configuration system internally.

9.2 Plotting with pandas and seaborn

matplotlib can be a fairly low-level tool. You assemble a plot from its base components: the data display (i.e., the type of plot: line, bar, box, scatter, contour, etc.), legend, title, tick labels, and other annotations.

In pandas we may have multiple columns of data, along with row and column labels. pandas itself has built-in methods that simplify creating visualizations from DataFrame and Series objects. Another library is `seaborn`, a statistical graphics library created by Michael Waskom. Seaborn simplifies creating many common visualization types.



Importing seaborn modifies the default matplotlib color schemes and plot styles to improve readability and aesthetics. Even if you do not use the seaborn API, you may prefer to import seaborn as a simple way to improve the visual aesthetics of general matplotlib plots.

Line Plots

Series and DataFrame each have a `plot` attribute for making some basic plot types. By default, `plot()` makes line plots (see Figure 9-13):

```
In [60]: s = pd.Series(np.random.randn(10).cumsum(), index=np.arange(0, 100, 10))

In [61]: s.plot()
```

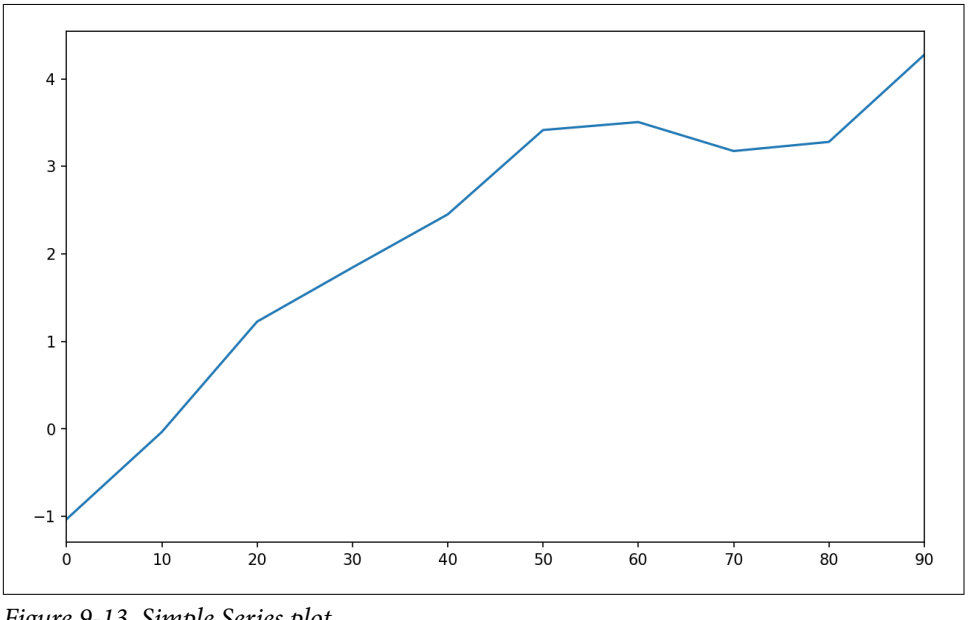


Figure 9-13. Simple Series plot

The Series object's index is passed to matplotlib for plotting on the x-axis, though you can disable this by passing `use_index=False`. The x-axis ticks and limits can be adjusted with the `xticks` and `xlim` options, and y-axis respectively with `yticks` and

`ylim`. See Table 9-3 for a full listing of `plot` options. I'll comment on a few more of them throughout this section and leave the rest to you to explore.

Most of pandas's plotting methods accept an optional `ax` parameter, which can be a matplotlib subplot object. This gives you more flexible placement of subplots in a grid layout.

DataFrame's `plot` method plots each of its columns as a different line on the same subplot, creating a legend automatically (see Figure 9-14):

```
In [62]: df = pd.DataFrame(np.random.randn(10, 4).cumsum(0),
   ....:                   columns=['A', 'B', 'C', 'D'],
   ....:                   index=np.arange(0, 100, 10))

In [63]: df.plot()
```

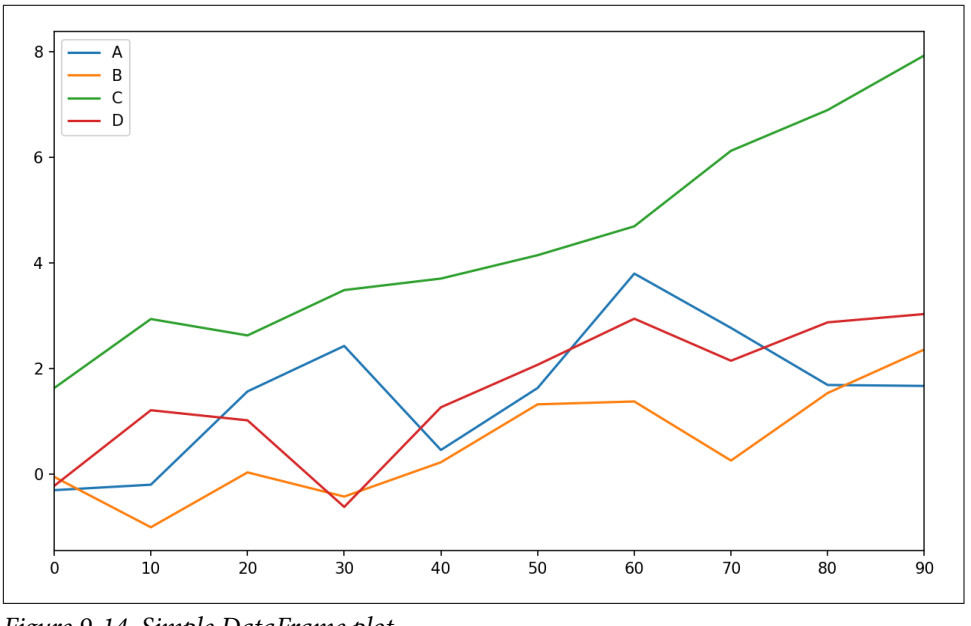


Figure 9-14. Simple DataFrame plot

The `plot` attribute contains a "family" of methods for different plot types. For example, `df.plot()` is equivalent to `df.plot.line()`. We'll explore some of these methods next.



Additional keyword arguments to `plot` are passed through to the respective matplotlib plotting function, so you can further customize these plots by learning more about the matplotlib API.

Table 9-3. Series.plot method arguments

Argument	Description
`label`	Label for plot legend
`ax`	matplotlib subplot object to plot on; if nothing passed, uses active matplotlib subplot
`style`	Style string, like `'ko--'`, to be passed to matplotlib
`alpha`	The plot fill opacity (from 0 to 1)
`kind`	Can be `'area'`, `'bar'`, `'barh'`, `'density'`, `'hist'`, `'kde'`, `'line'`, `'pie'`
`logy`	Use logarithmic scaling on the y-axis
`use_index`	Use the object index for tick labels
`rot`	Rotation of tick labels (0 through 360)
`xticks`	Values to use for x-axis ticks
`yticks`	Values to use for y-axis ticks
`xlim`	x-axis limits (e.g., `[0, 10]`)
`ylim`	y-axis limits
`grid`	Display axis grid (on by default)

DataFrame has a number of options allowing some flexibility with how the columns are handled; for example, whether to plot them all on the same subplot or to create separate subplots. See Table 9-4 for more on these.

Table 9-4. DataFrame-specific plot arguments

Argument	Description
`subplots`	Plot each DataFrame column in a separate subplot
`sharex`	If `subplots=True`, share the same x-axis, linking ticks and limits
`sharey`	If `subplots=True`, share the same y-axis
`figsize`	Size of figure to create as tuple
`title`	Plot title as string
`legend`	Add a subplot legend (`True` by default)
`sort_columns`	Plot columns in alphabetical order; by default uses existing column order



For time series plotting, see Chapter 11.

Bar Plots

The `plot.bar()` and `plot.barh()` make vertical and horizontal bar plots, respectively. In this case, the Series or DataFrame index will be used as the x (`bar`) or y (`barh`) ticks (see Figure 9-15):

```
In [64]: fig, axes = plt.subplots(2, 1)

In [65]: data = pd.Series(np.random.rand(16), index=list('abcdefghijklmnop'))

In [66]: data.plot.bar(ax=axes[0], color='k', alpha=0.7)
Out[66]: <matplotlib.axes._subplots.AxesSubplot at 0x7fb62493d470>

In [67]: data.plot.barh(ax=axes[1], color='k', alpha=0.7)
```

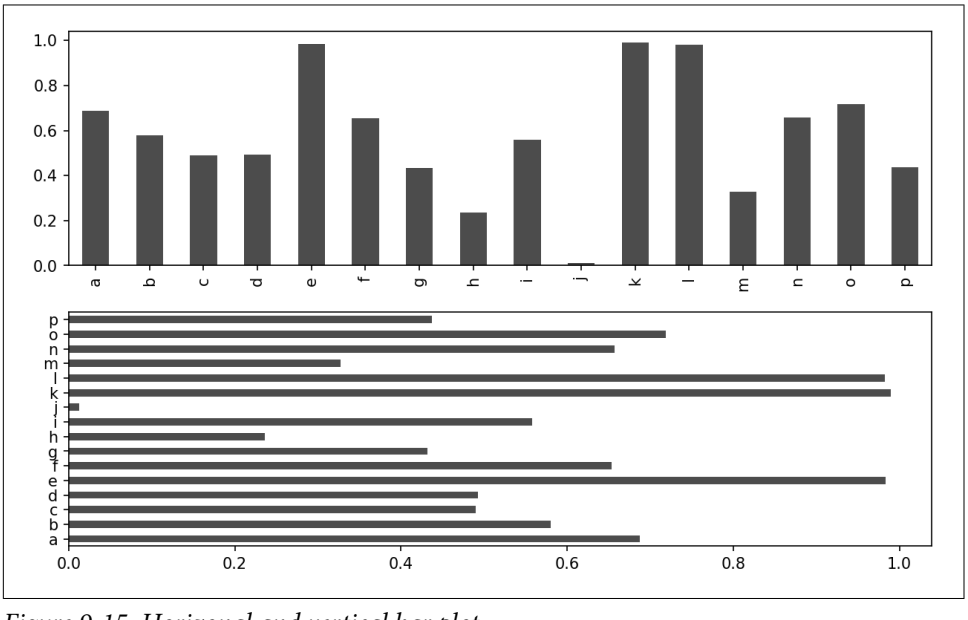


Figure 9-15. Horizonal and vertical bar plot

The options `color='k'` and `alpha=0.7` set the color of the plots to black and use partial transparency on the filling.